VOYAGE ANALYTICS

End-to-end architecture | Train once, serve many requests, improve through reviewed releases

A. MODEL DEVELOPMENT • Shared Python code, explained in three Colab notebooks

Course datasets
flights.csv • users.csv
hotels.csv
Link users and trips safely.

Validate & prepare
Check schema; explore data.
Split train / validation / test.
Fit preprocessing on training.

Train & evaluate
Flight-price regression
Educational label classification
Hotel recommendation ranking

Approve & save
Select using validation scores.
Evaluate on the held-out test.
joblib + version + checksum

Orchestrates regression pipeline

Log experiments

Reviewed handoff

B. AUTOMATION & RELEASE • Administrative services run locally

Airflow + PostgreSQL
Weekly regression DAG
Validate → train → approve
New candidate stays isolated.

MLflow tracking
Parameters, metrics, run IDs
Model versions and artifacts
Experiment record, not the API

GitHub release source
Application + shared code
Reviewed, approved bundles
Notebooks + infrastructure

Jenkins + Docker
Checkout → 13 tests → build
Image includes pinned models.
Run real prediction checks.

Local CD verified

C. RUNNING APPLICATION • A user request performs inference; it does not retrain models

User
Enter journey details
or hotel preferences.
Read the returned result.

Streamlit interface
Public: Community Cloud
Flights • hotels • insights
Educational classification lab

Flask prediction API
Validate incoming JSON.
Load saved models at startup.
Return result + model version.

Deployment environments
PUBLIC: Render Docker API
LOCAL: Jenkins Docker release
LOCAL: Kubernetes replicas
Kubernetes scaled manually.

Input → Streamlit → Flask → saved model → result on screen

Human review
informs retraining

MONITOR, REVIEW & IMPROVE
Request latency, errors and model version; historical drift reports.
Actual later outcomes are needed to measure prediction accuracy.
A reviewed retraining decision starts the next model-development cycle.

READ THE CONNECTIONS

Solid arrows: data / software / requests. Dashed arrows: orchestration, tracking or a reviewed handoff.

Render is a separate public deployment. Jenkins-to-Kubernetes CD and automatic HPA scaling are not verified.